

EX-NO:07**TEMPORAL DATABASE****10-11-23****AIM:**

To Create a Temporal Database and explore the usage of Temporal Queries.

VALID TIME:**CREATE TABLE:**

```
SQL> CREATE TABLE Location (Id INT PRIMARY KEY, Name VARCHAR2(15),  
City VARCHAR2(15), Valid_start TIMESTAMP, Valid_end TIMESTAMP);
```

```
Table created.
```

INSERT:

```
SQL> Insert into location values (1, 'Steven', 'Roma',  
TO_TIMESTAMP ('2023-01-03 09:12:00', 'YYYY-MM-DD HH24:MI: SS'),  
TO_TIMESTAMP ('2023-10-19 17:13:00', 'YYYY-MM-DD HH24:MI: SS'));
```

```
1 row created.
```

```
SQL> insert into location values (2, 'Lex', 'Tokyo',  
TO_TIMESTAMP ('2023-02-19 12:00:00', 'YYYY-MM-DD HH24:MI: SS'),  
NULL);
```

```
1 row created.
```

```
SQL> insert into location values (3, 'Bruce', 'Texas',  
TO_TIMESTAMP ('2023-01-14 14:20:00', 'YYYY-MM-DD HH24:MI: SS'),  
TO_TIMESTAMP ('2023-11-14 14:00:00', 'YYYY-MM-DD HH24:MI: SS'));
```

```
1 row created.
```

```
SQL> insert into location values (4,'Nancy', 'Toronto',
TO_TIMESTAMP ('2022-12-27 10:10:00', 'YYYY-MM-DD HH24:MI: SS'),
TO_TIMESTAMP ('2023-08-27 10:00:00', 'YYYY-MM-DD HH24:MI: SS'));
```

```
1 row created.
```

```
SQL> insert into location values (5, 'Ismael', 'Bombay',
TO_TIMESTAMP ('2022-12-31 01:00:00', 'YYYY-MM-DD HH24:MI: SS'),
SYSDATE);
```

```
1 row created.
```

DISPLAY THE INSERTED VALUES:

```
SQL> select * from location;
```

ID	NAME	CITY	VALID_START	VALID_END
1	Steven	Roma	03-JAN-23 09.12.00.000000 AM	19-OCT-23 05.13.00.000000 PM
2	Lex	Tokyo	19-FEB-23 12.00.00.000000 PM	
3	Bruce	Texas	14-JAN-23 02.20.00.000000 PM	14-NOV-23 02.00.00.000000 PM
4	Nancy	Toronto	27-DEC-22 10.10.00.000000 AM	27-AUG-23 10.00.00.000000 AM
5	Ismael	Bombay	31-DEC-22 01.00.00.000000 AM	21-NOV-23 02.14.57.000000 PM

RETRIEVING LOCATIONS VALID AT A SPECIFIC DATE AND TIME:

```
SQL> select * from location where valid_start <= TIMESTAMP '2023-08-15 14:00:00'
and valid_end is NULL;
```

ID	NAME	CITY	VALID_START	VALID_END
2	Lex	Tokyo	19-FEB-23 12.00.00.000000 PM	

```
SQL> select * from location where valid_start >= TIMESTAMP '2022-10-15 14:00:00'
and valid_end is not null;
```

ID	NAME	CITY	VALID_START	VALID_END
1	Steven	Roma	03-JAN-23 09.12.00.000000 AM	19-OCT-23 05.13.00.000000 PM
3	Bruce	Texas	14-JAN-23 02.20.00.000000 PM	14-NOV-23 02.00.00.000000 PM
4	Nancy	Toronto	27-DEC-22 10.10.00.000000 AM	27-AUG-23 10.00.00.000000 AM
5	Ismael	Bombay	31-DEC-22 01.00.00.000000 AM	21-NOV-23 02.14.57.000000 PM

UPDATE:

SQL> update location set

valid_start= TO_TIMESTAMP ('14-01-2022 14:00:00', 'DD-MM-YYYY HH24:MI: SS')

where id=3;

1 row updated.

SQL> select * from location where valid_start <= TIMESTAMP '2022-01-15 14:00:00'

and valid_end >= TIMESTAMP '2023-01-01 14:00:00';

ID	NAME	CITY	VALID_START	VALID_END
3	Bruce	Texas	14-JAN-22 02.00.00.000000 PM	14-NOV-23 02.00.00.000000 PM

FINDING LOCATIONS VALID TODAY:

SQL> select * from location where valid_start <= TRUNC(SYSDATE)

and valid_end is NULL or valid_end >= TRUNC(SYSDATE);

ID	NAME	CITY	VALID_START	VALID_END
2	Lex	Tokyo	19-FEB-23 12.00.00.000000 PM	
5	Ismael	Bombay	31-DEC-22 01.00.00.000000 AM	21-NOV-23 02.14.57.000000 PM

FINDING EXPIRED LOCATIONS:

SQL> select * from location where valid_end is not null and valid_end < SYSTIMESTAMP;

ID	NAME	CITY	VALID_START	VALID_END
1	Steven	Roma	03-JAN-23 09.12.00.000000 AM	19-OCT-23 05.13.00.000000 PM
3	Bruce	Texas	14-JAN-22 02.00.00.000000 PM	14-NOV-23 02.00.00.000000 PM
4	Nancy	Toronto	27-DEC-22 10.10.00.000000 AM	27-AUG-23 10.00.00.000000 AM
5	Ismael	Bombay	31-DEC-22 01.00.00.000000 AM	21-NOV-23 02.14.57.000000 PM

GREATER THAN:

SQL> select * from location where valid_start > TIMESTAMP '2023-01-01 00:00:00';

ID	NAME	CITY	VALID_START	VALID_END
1	Steven	Roma	03-JAN-23 09.12.00.000000 AM	19-OCT-23 05.13.00.000000 PM
2	Lex	Tokyo	19-FEB-23 12.00.00.000000 PM	

AGGREGATE FUNCTION:**COUNT:**

SQL> select COUNT (*) as total_rows from location;

```
TOTAL_ROWS
-----
          5
```

MIN & MAX:

SQL> select MIN(valid_start) As earliest_start, MAX(Valid_start) As Latest_start from location;

```
EARLIEST_START                LATEST_START
-----
14-JAN-22 02.00.00.000000 PM  19-FEB-23 12.00.00.000000 PM
```

AVG:

SQL> select AVG (EXTRACT (DAY FROM (valid_end - valid_start)) *24*60*60
+ EXTRACT (HOUR FROM (valid_end - valid_start)) *60*60
+ EXTRACT (MINUTE FROM (valid_end - valid_start)) *60
+ EXTRACT (SECOND FROM (valid_end - valid_start))) / (24*60*60)
as avg_duration_days from location;

```
AVG_DURATION_DAYS
-----
      381.719783
```

UPDATE:

SQL> update location set valid_end = TO_TIMESTAMP('2023-11-04 00:00:00',
'YYYY-MM-DD HH24:MI:SS') where id=2;

```
1 row updated.
```

SQL> select * from location;

ID	NAME	CITY	VALID_START	VALID_END
1	Steven	Roma	03-JAN-23 09.12.00.000000 AM	19-OCT-23 05.13.00.000000 PM
2	Lex	Tokyo	19-FEB-23 12.00.00.000000 PM	04-NOV-23 12.00.00.000000 AM
3	Bruce	Texas	14-JAN-22 02.00.00.000000 PM	14-NOV-23 02.00.00.000000 PM
4	Nancy	Toronto	27-DEC-22 10.10.00.000000 AM	27-AUG-23 10.00.00.000000 AM
5	Ismael	Bombay	31-DEC-22 01.00.00.000000 AM	21-NOV-23 02.14.57.000000 PM

ADDING TRANSACTION TABLE:**ALTER:**

SQL> Alter table location add(Transaction_start DATE, Transaction_end DATE);

Table altered.

UPDATING TRANSACTION TIME:

SQL> update location set

valid_end = TO_TIMESTAMP ('2023-05-16 05:13:00', 'YYYY-MM-DD HH24:MI: SS'),
 transaction_start = TO_DATE ('2023-05-16', 'YYYY-MM-DD'),
 transaction_end = SYSDATE where id=1;

1 row updated.

SQL> update location set

valid_end = TO_TIMESTAMP ('2023-04-16 05:13:00', 'YYYY-MM-DD HH24:MI: SS'),
 transaction_start = TO_DATE ('2023-04-16', 'YYYY-MM-DD'),
 transaction_end = SYSDATE where id=2;

1 row updated.

SQL> update location set

valid_end = TO_TIMESTAMP ('2023-06-16 05:13:00', 'YYYY-MM-DD HH24:MI: SS'),
 transaction_start = TO_DATE ('2023-06-16', 'YYYY-MM-DD'),
 transaction_end = SYSDATE where id=3;

1 row updated.

SQL> update location set

transaction_start = TO_DATE ('2023-08-23', 'YYYY-MM-DD'),
 transaction_end = SYSDATE where id=4;

1 row updated.

SQL> update location set

valid_end = TO_TIMESTAMP ('2023-07-16 05:13:00', 'YYYY-MM-DD HH24:MI: SS'),

transaction_start = TO_DATE ('2023-07-16', 'YYYY-MM-DD'),

transaction_end = SYSDATE where id=5;

1 row updated.

DISPLAYING THE TRANSACTION TABLE AFTER UPDATING:

SQL> select * from location;

ID	NAME	CITY	VALID_START	VALID_END	TRANSACTION_START	TRANSACTION_END
1	Steven	Roma	03-JAN-23 09.12.00.000000 AM	16-MAY-23 05.13.00.000000 AM	16-MAY-23	22-NOV-23
2	Lex	Tokyo	19-FEB-23 12.00.00.000000 PM	16-APR-23 05.13.00.000000 AM	16-APR-23	22-NOV-23
3	Bruce	Texas	14-JAN-22 02.00.00.000000 PM	16-JUN-23 05.13.00.000000 AM	16-JUN-23	22-NOV-23
4	Nancy	Toronto	27-DEC-22 10.10.00.000000 AM	16-MAY-23 05.13.00.000000 AM	23-AUG-23	22-NOV-23
5	Ismael	Bombay	31-DEC-22 01.00.00.000000 AM	16-JUL-23 05.13.00.000000 AM	16-JUL-23	22-NOV-23

RETRIEVING LOCATIONS TRANSACTION AT A SPECIFIC DATE:

SQL> select * from location where transaction_start > DATE '2023-04-30';

ID	NAME	CITY	VALID_START	VALID_END	TRANSACTION_START	TRANSACTION_END
1	Steven	Roma	03-JAN-23 09.12.00.000000 AM	16-MAY-23 05.13.00.000000 AM	16-MAY-23	22-NOV-23
3	Bruce	Texas	14-JAN-22 02.00.00.000000 PM	16-JUN-23 05.13.00.000000 AM	16-JUN-23	22-NOV-23
4	Nancy	Toronto	27-DEC-22 10.10.00.000000 AM	16-MAY-23 05.13.00.000000 AM	23-AUG-23	22-NOV-23
5	Ismael	Bombay	31-DEC-22 01.00.00.000000 AM	16-JUL-23 05.13.00.000000 AM	16-JUL-23	22-NOV-23

SQL> select * from location where transaction_start < DATE '2023-04-30' and
transaction_end <= SYSDATE;

ID	NAME	CITY	VALID_START	VALID_END	TRANSACTION_START	TRANSACTION_END
2	Lex	Tokyo	19-FEB-23 12.00.00.000000 PM	16-APR-23 05.13.00.000000 AM	16-APR-23	22-NOV-23

SQL> select * from location where transaction_end is not null and

transaction_end < SYSDATE;

ID	NAME	CITY	VALID_START	VALID_END	TRANSACTION_START	TRANSACTION_END
1	Steven	Roma	03-JAN-23 09.12.00.000000 AM	16-MAY-23 05.13.00.000000 AM	16-MAY-23	22-NOV-23
2	Lex	Tokyo	19-FEB-23 12.00.00.000000 PM	16-APR-23 05.13.00.000000 AM	16-APR-23	22-NOV-23
3	Bruce	Texas	14-JAN-22 02.00.00.000000 PM	16-JUN-23 05.13.00.000000 AM	16-JUN-23	22-NOV-23
4	Nancy	Toronto	27-DEC-22 10.10.00.000000 AM	16-MAY-23 05.13.00.000000 AM	23-AUG-23	22-NOV-23
5	Ismael	Bombay	31-DEC-22 01.00.00.000000 AM	16-JUL-23 05.13.00.000000 AM	16-JUL-23	22-NOV-23

RESULT:

Thus, created a database using temporal database and performed the temporal queries successfully.

OBSERVATION	RECORD	SIGNATURE